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Paper 1 of 1 — Acc 410 Forensic Accounting a n d F r a u d

Level 400 • Second Semester • Handout

Lecture Notes on Forensic Accounting a n d F r a u d

E x a m i n a t i o n s (ACC 410)

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PREAMBLES:

To date, there are no comprehensive authoritative standards or guidelines that can strictly apply to the education and practice of forensic accounting. Because of an almost limitless variety of areas of practice in forensic accounting, it may be difficult to develop comprehensive course outlines and detailed guidance in this area.

1. INTRODUCTION TO FORENSIC ACCOUNTING

1.1 Forensic Accounting Education: Evolution, Need and Benefits

Evolution of forensic accounting education availability

Over a decade ago, leaders in the accounting profession began to call for educators to provide forensic accounting education, owing in part to widely publicized financial reporting scandals and the scarcity of schools providing this education (Melancon, 2002). Academia had largely ignored the field. Academic articles focusing on forensic accounting education were rare prior to the late 1990s (Rebele et al., 1998). One of the earliest studies, Rezaee et al. (1996), identified only four universities that offered a separate forensic accounting course. Buckhoff and Schrader (2000) found 13 universities offered a separate forensic accounting course, consistent with what was found by another independent study (Peterson and Reider, 1999). Combining those two studies, Peterson and Reider (2001) identified 19 universities offering a forensic accounting course, reviewed the syllabi for similarities and differences in content, and learning activities. Carnes and Gierlasinski (2001) reported that universities were slow in providing forensic accounting education to students, even though the changing accounting environment was demanding students be taught these skills. Bundy et al. (2003) noted that forensic accounting was a new growth industry in accounting, and found only two US colleges offered an undergraduate degree in a related field: economic crime investigation. Rezaee et al. (2004)

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found small improvement in the availability of forensic accounting education, identifying 25 universities that offered a separate forensic accounting course. At approximately this same time, in 2003, West Virginia University (WVU) received a significant grant from the National Institute of Justice to develop a model curriculum in forensic accounting in collaboration with a technical working group comprising 46 subjectmatter experts from a variety of stakeholder groups (WVU, 2007). WVU developed the grant proposal because only a few schools were offering multiple courses in the forensic accounting field, with that variation in course content. In a review of

accounting education literature published in six academic journals over four years, Apostolou et al. (2010, p. 181) note, “[. . .]in 2006-2009, an increase in articles related to fraud and forensic issues and ethics reflects the changing emphasis on these topics”. Articles began to appear that recommended forensic accounting subject matter content, such as behavioral sciences (Ramamoorti, 2008), ethics and legal/regulatory environments (Curtis, 2008), digital forensics (Pearson and Singleton, 2008) and litigation advisory services (Heitger and Heitger, 2008).

Later studies generally found an increase in the number of colleges and universities offering forensic accounting education. For example, Seda and Kramer (2009) identified 48 universities and colleges that offered a separate forensic accounting course and found another 21 institutions with various forensic accounting programs at the undergraduate and graduate levels (degrees, minors and/or certificates). Smith and Crumbley (2009) found 40 universities with one to six forensic accounting courses offered.

Need and benefits of Forensic Accounting Education

International fraud and economic crime surveys consistently report that fraud is a serious problem affecting organizations, without regard to geographical location, size or industry (ACFE, 2014; E&Y, 2014; PwC, 2014; KPMG, 2013). A constant finding is the most common way of discovering a fraud is through a tip, regardless of the source (e.g. employee, vendor, customer or anonymous). Further, the surveys report that external auditors rarely discover a fraud because many times it may not be causing financial statements to be materially misstated. Owing to the significant amount of employee and management fraud, embezzlement and other financial crimes occurring in today’s society, accounting and auditing professionals must have appropriate forensic accounting training and skills to recognize those crimes (Houck et al., 2006). These facts underscore the necessity for courses specifically focused on forensic accounting, rather than assuming auditing courses provide sufficient training.

Beyond benefiting society by providing forensic accounting education, students who acquire this knowledge directly benefit. Empirical evidence has been found that hiring potential is enhanced when a candidate has completed a forensic accounting course (Mounce and Frazier, 2002). As far back as 1991, the national placement director for one of the largest recruiters of accountants, financial personnel and information specialists stated that forensic accounting “is the hottest

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field in accounting firms” with salaries ranging from \$50,000 to \$200,000 (Fowler, 1991). A few years later, an article in The Wall Street Journal labeled forensic accountants as the accounting profession’s “new glamour kids” and reported that not only did the Federal Bureau of Investigation (FBI) double its forensic accountant agent force between 1992 and 1996, but several large public accounting firms doubled or even quintupled their forensic ranks (MacDonald, 1996). A cover story in USA News & World Report included forensic accountant as one of the eight most secure career tracks in the country (Levine, 2002). The article reported that 40 per cent of the top 100 accounting firms were expanding their forensic accounting consulting services. Demand for forensic accountants remains strong today and continues to grow (Cohn, 2014; White, 2013; Tysiac, 2012). Employment opportunities are abundant and include public accounting, loss

prevention/security, internal audit, law enforcement and a number of federal agencies, such as the FBI and the Internal Revenue Service. Even if students never choose a forensic accounting career path, the skills they learn will make them better business professionals, whether they work as an auditor, manager or consultant. For example, Carpenter et al. (2011) provide empirical evidence that students who completed a forensic accounting course possess more professional skepticism than students without such training, and that this effect persists over time.

There are no comprehensive standards or guidelines for education and practice of forensic accounting

There are many benefits for accountants who engage in forensic accounting practices. The first is the increase in income and high per-hour yield. Because there are comparably fewer accountants practising forensic accounting, than in other fields of accounting, and such a great demand for these services, forensic accountants with expertise in the field can demand substantial hourly rates.

Furthermore, court cases do not follow the predictable "feast and famine" cycle of an accounting firm's usual workload. Thus, this very desirable income can be received during slow times of the year, not only balancing the yearly workload but allowing for a more steady level of cash flow. Second, forensic accounting practices are not only exciting and challenging but also allow accountants to exercise skills like innovation, creativity and imagination, that go beyond the more routine work of an accountant and accordingly permit an increase in personal and professional satisfaction. Finally, for small to medium-sized firms, a well-established practice in forensic accounting may allow a firm to obtain the market "niche" as an expert in the field. This market niche may be a key to improvement in a firm's current practice as well as a stepping-stone for future practice development.

Offsetting the above benefits are some of the difficulties and problems associated with practising forensic accounting. First, forensic accountants must be more knowledgeable about such things as law, courtroom procedures, and effective testimony. Development of such skills requires a substantial commitment of time, effort and resources to train and educate accountants about this new practice. Second, the skills, such as innovation, creativity and ingenuity, required for success as a forensic accountant are not usually found among the less experienced members of an accounting firm. Thus, forensic accounting practices may put a great amount of work on the senior members of a firm. Finally, the promotional and marketing effort required to develop a forensic accounting practice may be quite extensive with the pay-off not being seen for many years. Some accounting firms may not have the resources available to expand their efforts in such a manner.

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The problems and obstacles in the way of integration of forensic accounting into the accounting curriculum can be summarized as follows. The first problem is the lack of faculty and administration interest. Academicians may not be convinced that they can make a great contribution to the breadth of accounting students' education by integrating forensic accounting into the accounting curriculum. The second problem pertains to the lack of authoritative guidance for faculty in teaching forensic accounting.

To date, there are no comprehensive authoritative standards or guidelines that can strictly apply to

the education and practice of forensic accounting. Because of an almost limitless variety of areas of practice in forensic accounting, it may be difficult to develop comprehensive course outlines and detailed guidance in this area. The third problem is the lack of the required knowledge in forensic accounting on the part of the faculty member. In teaching a forensic accounting course, the faculty member should understand not only the forensic accounting techniques and procedures but also the industry itself. The final and perhaps the most important problem is in obtaining sufficient financial resources; faculty members are often faced with limited resources at universities. This problem can be resolved by requesting financial support from public accounting firms and industry.

FORENSIC ACCOUNTING IN THE CURRICULUM BY Zabihollah Rezaee, Gerald H.

Lander and Thomas A. Gavin, *Managerial Auditing Journal*, Vol. 7 No. 3, 1992, pp. 25-29

Recent History

Maurice E. Peloubet is credited with developing the term Forensic Accounting in his 1946 essay "Forensic Accounting: Its Place in Today's Economy." By this time, Forensic Accounting had proven its worth during World War II, however formalized procedures were not in place until the 1980's when major academic works were published. During the O.J. Simpson trial, forensic accountants evaluated his assets impacting the damages awarded in the civil suit. Forensic Accounting has been pivotal in the corporate scandals of companies such as Enron, Tyco and Worldcom.

Professional History

In 1992 the American College of Forensic Examiners was established with the American Board of Forensic Accounts starting in 1997. The *Journal of Forensic Accounting, Auditing, Fraud and Taxation* began publication in 2000. In 2002 the Sarbanes-Oxley Act established the Public Companies Accounting Oversight Board (PCAOB) which was charged with developing auditing standards, conducting investigations and ensuring corporate compliance. As a result of the Sarbanes-Oxley Act, there has been continuing emphasis on forensic accounting.

Forensic accountants are Chartered Accountants/Certified Public Accountants who have obtained additional certification as a forensic accountant or as a fraud examiner. They work in several different venues. Fraud detection and prevention is the largest field, but forensic accounting is also used in civil and criminal litigation such as divorces or bankruptcy, government work and even in detecting the financial footprint of terrorism.

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1.2 Definition of Forensic Accounting, Litigation Accounting, and Investigative Accounting.

There are a variety of definitions that attempt to describe the range of forensic services offered within accounting profession. Internationally, the ACFE (2011) define it as the use of professional accounting skills in matters involving potential or actual civil or criminal litigation. From an Australian perspective, the Accounting Professional and Ethical Standards Board (APESB)[3], working with FASIG and other interested parties, issued APES 215 forensic accounting services[4] – which defines forensic accounting services as meaning “expert witness

services, lay witness services, consulting expert services investigation services” (APESB, 2008, p. 3) Akkeren, 2013 Vol. 25 No. 2, 2013 pp. 188-216

Bolgna and Linqvist (1995) defined forensic accounting as the application of financial skills and an investigative mentality to unresolved issues, conducted within the context of the rules of evidence of a particular country. According to Digabriele (2008; 2009), forensic accountants should possess various combinations of skill and knowledge in accounting, auditing, law, and investigation techniques. These should be accompanied by strong ethical values and soft skills. Although the main thrust of forensic accounting is involved with the financial aspects of an investigation, it encompasses all the necessary investigative expertise and experience such as interrogative skills, knowledge of law and rules of evidences, investigative proficiency, and interpersonal skills (Syed Ahmad, 2008).

Forensic auditing is focused on the identification, interpretation, and communication of the evidence of underlying strategic economic and reporting events. It not single-event based, like a fraud examination, and a forensic audit is not used to render an audit opinion. As such, forensic audits are easily adapted to a principles-based accounting environment with broad guidelines applied to a variety of accounting investigations without using rule-based audit approaches or more narrowly-focused fraud practices.

Major corporate scandals such as Enron, WorldCom, and Olympus, to name a few, have raised concern about the effectiveness of traditional financial audit to provide oversight of financial activities.

According to Oyedokun (2013), Forensic Accounting is a Scientific Accounting method of uncovering, resolving, analyzing and presenting fraud matters in a manner that is acceptable in the court of law.

There have been differences in the definition of the fraud related audit. The terms “forensic audit”, “forensic accounting”, “fraud examination” “financial forensics” “ Litigation accounting”

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and “Investigative Accounting” to name a few are often used to describe processes that are similar in some cases but are different in others. The Founder of the ACFE, Joseph T. Wells, for example, argues that (Wells, 2005, p. 4):

Resolving allegations of fraud – from tips, complaints, or accounting clues – is the discipline of fraud examination. It involves obtaining documentary evidence, interviewing witnesses and potential suspects, writing investigative reports, testifying to findings, and assisting in the general detection and prevention of fraud. Fraud examination has similarities to the field of forensic accounting, but the two terms are not precisely the same. The latter is the use of any accounting knowledge or skill for courtroom purposes. Forensic accounting can therefore involve not only fraud, but also bankruptcy, business valuations and disputes, divorce, and a host of other litigation support services. Conversely, although accountants typically perform fraud examinations, professionals in other fields, such as law enforcement officials, corporate security specialists, or private investigators, can also conduct them.

Despite the differences in how process is being defined across literatures, essentially, every fraud investigation is a process of finding the answers to a set of predetermined questions based on facts and evidences and conducted using legally accepted methods. A forensic accountant needs to be able recognize early indications of fraud from the “red flags” such as excessive life styles, accounting anomalies, and unusually close relationship with vendors and clearly determines what is it that he must investigate. In other words, he needs to be able to formulate the so-called “investigation questions” to guide him throughout the course of the investigation. Such questions will keep a forensic accountant from straying from the goals of the investigation. Unlike the financial auditors, forensic accountants do not have the luxury of an agreed upon “technical standard” and thus their ultimate tool in performing their tasks is no other than their own logic. When presenting their findings in the court or in front of the people who commissioned the investigation, for example, a forensic accountant needs to be able to explain the entire course of his or her fraud investigation in a logical way that will make sense even to laymen. Forensic accounting, sometimes referred to as investigative accounting, is a unique career field that combines accounting with information technology.

A Forensic Accountant uses his knowledge of accounting, law, investigative auditing and criminology to uncover fraud, find evidence and present such evidence in court if required to. In the same vein, Investigative accounting is often associated with investigation of criminal matters. A typical investigative accounting assignment will be an investigation of employee theft. Other example includes securities fraud, kickbacks and proceeds of crime investigations.

Litigation Consultant/Accountant

Litigation consultants participate in a legal action by assisting lawyers to gather, detect and analyse

data, particularly financial data. Data gathering and detection is the process of obtaining data and determining the relevance, adequacy and accuracy of information pertinent to a legal action. Data analysis relates to the procedures for evaluating and ascertaining information presented to the trier of fact (e.g. the jury, the court) and educating the attorneys about data collection and valuation procedures as well as the financial interpretations of the result of the analysis. Data analysis is

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often one of the most important duties of a forensic appraiser, because the outcome of the testimony depends on the presentation made by the expert witness.

Preparing for Litigation

You might have heard the phrase “forensic evidence” before, which simply means evidence that is able to be presented in a court of law. Hence, forensic accounting is a term to describe an analysis of financial information that can be used to support a case in a court of law. The process of digging through all of a company’s or individual’s financial information can take months or even years and requires a team of specialized accountants that act like detectives trying to solve a mystery.

Typically, an accounting firm will be engaged by a client either looking to defend themselves, or one looking to prosecute someone. Most medium- to large-sized firms have a forensic accounting

department, which may consist of various forensic auditors.

Litigation supports – it deals primarily with the issues related to the quantification of economic damages. A typical litigation support assignment would be calculating the economic loss resulting from a breach of contract. Some of the types of litigation cases where the forensic accountant can be of assistance include shareholder/partner disputes, matrimonial etc.

ACCOUNTANTS AS EXPERT WITNESSES

Accountants may appear in court in one of two situations: when giving evidence of fact and when acting as expert witnesses in giving opinion evidence. This important distinction between the two situations is explained below.

Specialist as witness of fact

Forensic accountants can act as witnesses of fact where they can state as a fact, from their own knowledge and examination of a particular item, that a certain event or sequence of events occurred in a particular instance. Generally speaking, an ordinary witness providing testimony in court can only give evidence of fact, e.g. what they saw, heard, tasted, smelt, touched – facts collected by one of the five senses. Accountants acting as witnesses of fact should provide evidence in court based on what happened, not on supposition. They are not permitted to speculate or to give an opinion as to the circumstances leading up to an event they have witnessed.

Specialist as expert witness who gives evidence of opinion

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An expert witness is a person qualified, by virtue of special knowledge, skill, experience or training to help present and reach conclusions of fact in areas beyond the experience and comprehension of an ordinary person. The court has accepted accountants as experts to give testimony in the form of an opinion based on facts made known to accountants before the trial. The accountant should gather sufficient and competent evidence in forming an opinion as an expert witness. The expert witness may testify as to his or her opinion or answer hypothetical questions as well as relate facts directly known to him or her. Although there is no standard or minimum level of expertise required, the Rule 702 of the Federal Rules of Evidence requires that the judge rules that the proposed expert possesses sufficient skill, knowledge or expertise in assisting the trier of fact in determining the facts of the case.

The expert witness's testimony, presentation, and behaviour during cross-examination, are crucial to the outcome of the case. Typically, opposing lawyers will do everything possible to discredit the witness's testimony by challenging their knowledge and analysis of the information available. An accountant serving as an expert.

It should be noted that, the expert's primary obligation is to the court not the hiring party. Expert evidence is not a substitute for the exercise of the court's own judgement

Forensic accountants also act as expert witnesses and in that role they can, from their own knowledge and experience, give an opinion as to why a certain event or sequence of events did or did not occur. The trial judge will decide when a witness is allowed to give

expert opinion. Expert witnesses may be called upon to state their opinion on a matter within their specialised knowledge and skill where the court itself cannot form an opinion from the facts. Forensic accountants often present highly technical material to courts in terminology they can understand.

Fraud Examiner

More and more cases of fraud are coming to the public's attention. For example, the government estimates that 70 per cent of the recent failure in the savings and loan associations involved a degree of fraud[6]. The pervasiveness of this issue has led to the creation of the National Commission on Fraudulent Financial Reporting in 1985[7]. The Commission was formed to examine the many complex issues surrounding the incidence of fraudulent reporting and to make recommendations to reduce their occurrence. The Commission recognized that accountants play a crucial role in detecting and deterring fraudulent financial reporting and, therefore, encouraged accountants to become experts in fraud auditing.

1.3 What does a Forensic Accountant do?

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The forensic accountant can be considered as a bloodhound (detective) of bookkeeping because he is able to sniff out fraud transactions, seeks for concrete evidence, find out the misstatement and look beyond the numbers (De Lorenzo, 1993). Thus, according to Albrecht et al. (2009), the most important skills that the forensic accountant should have are analytical skills, communication skills, technology skills, knowledge in accounting, business, and law and human behaviour. Forensic accountants utilize an understanding of business information and financial reporting systems, accounting and auditing standards and procedures, evidence gathering and investigative techniques, litigation processes and procedures to perform their work (Crumbley and Smith, 2009). The skills and preparation for precision, attention to detail, objectivity, problem-solving ability, and strong oral and written communication skills are important for forensic specialists (Torpe, 2009).,,,,,,,,,,,,,,,,,ETHNIC IN FORENSIC ACCOUNTING....

1.4 Types of Assignments a Forensic Accountant Perform (i.e. Forensic Accounting Techniques).

In the private sector, forensic accounting practices can be classified into three areas: fraud examiner; litigation consultant; and expert witness. In general, forensic accountants can practice as witnesses in criminal, tax matters, contract interpretation, insurance claims litigation, divorce and equitable property settlements, prevention and detection of fraud as well as vulnerability assessment both in the private and public sectors.

However, there are various type of forensic auditing that can take place, and the types of legal proceedings that they fall typically group them. Below are some of the most common examples:

- Financial theft (customers, employee or outsiders)

- Security fraud
- Bankruptcy
- Defaulting on debt
- Economic damages (various type of lawsuits to cover damages)
- Tax evasion or fraud
- Corporate valuation disputes
- Professional negligence claims
- Money laundering
- Privacy information
- Divorce proceedings STOPPED

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1.5 Characteristics and Qualities of a Forensic Accountant.

SKILLS FOR FORENSIC ACCOUNTANTS

Digabriele (2008) has identified and suggested the relevant skills for forensic accountants based on a nationwide survey from a random sample of 1,500 accounting academics, forensic accounting practitioners and users of forensic accounting services. The relevant skills of forensic accountants are 1) deductive analysis, 2) critical thinking, 3) unstructured problem solving, 4) investigative flexibility, 5) analytical proficiency 6) oral communication 7) written communication, 8) specific legal knowledge and 9) composure.

Another study by American Institute of Certified Public Accountants (AICPA report) found that communication skills, the ability to simplify the complex and the ability to present opinions in a legal setting are critical to the effectiveness of the forensic accountants (Davies, Farrell and Ogilby, 2010). The results obtained from the attorney, academicians and CPA (practitioners) reveal that analytical characteristics remain the foremost trait that forensic accountants are expected to possess. The results also reveal that communication skills, the ability to simplify the complex and the ability to present legal opinions in a legal setting are critical to the effectiveness of the forensic accountants.

Forensic accountant - applies specialized knowledge of fraud detection, prevention, deterrence and remediation structures to the fundamental knowledge in gathering information, investigation, analyzing, reporting and communicating financial information to improve future task performance judgment or to resolve legal matters (Popoola, 2014). Specifically, forensic accountants will continue to be in high demand (Wells, 2005) as long as criminals exist in the areas of fraud, corruption, asset misappropriation and conversion. This study aligns with Wells (2005) position that as long as untrained graduates are employed to detect fraud committed by technologically advanced criminals, the necessity for forensic accountants with fraud knowledge will continue to be on the increase.

In practice, according to this study's literature review and in-depth interviews with a number of

forensic accountants in Indonesia and the USA, three important ingredients are needed to create a forensic accountant: mentality, method, and experience. The first factor, mentality, is possibly the most difficult to cultivate. Mentality concerns one's way of thinking in performing forensic accounting tasks. This is built upon fundamental elements such as the ability to differentiate the

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right from the wrong, courage to stand up for what is right, ability to withstand the pressures from the works, and puzzle solving mindset. Second, a forensic accountant needs to understand various methods used in fraud investigation. Such knowledge can be acquired through education and training conducted by universities or other institutions. A forensic accountant also needs to have sufficient experience in fraud investigation in which he or she will have a chance to utilize the knowledge previously acquired through education and training. Ability to differentiate the right from the wrong To build an understanding of the "right and wrong" paradigm, one may start from understanding the foundation of morality which is often used as a standard to judge the righteousness of an action. In reality, however, such standard can be differently constructed across societies, which also mean that an action may be right from the perspective of one existing morality system but wrong from another (Ta'nnso, 2007, p. 130).

Literatures have long been debating about what causes such variations of morality standards. Many believe that culture plays an essential role in shaping the morality standard which also means that each culture has its own set of values which influence moral decision making and may include from laws and regulations to the notions of pride, honor, and justice (Tucker, 2009, p. 21). There are, however, those who believe that at least parts of our moral system have been coded into our genes since we were born (Hales, 2009, p. 234; Tucker, 2009, p. 23). There are also others who believe that there is a grammar - like mechanism which guides humans in their moral decisions. Hauser (2006), for example, argues that: [. . .] our moral faculty is equipped with a universal moral grammar, a toolkit for building specific moral systems. Once we have acquired our culture's specific moral norms - a process that is more like growing a limb than sitting in Sunday school and learning about vices and virtues - we judge whether actions are permissible, obligatory, or forbidden, without conscious reasoning and without explicit access to the underlying principles. A forensic accountant does not need to be drowned in the seemingly endless debates on why moral standards are what they are and what causes the variations as he or she only needs to know which moral system is accepted in a particular area. In practice, understanding of moral standard is not going to be enough to be used in a fraud investigation to decide whether an act is fraud or not. According to Miftah Muttaqin, a senior investigator at the Indonesian Corruption Eradication Commission (KPK), in a discussion with the researcher, a forensic accountant needs be able to recognize the boundaries between fraudulent and non-fraudulent acts in a particular environment through an understanding of the existing regulatory framework (e.g. criminal codes, business law, and professional code of conducts) within which he or she works (Muttaqin, 2013). Such framework usually sets the baseline definition for fraudulent acts. As Cressey (1950) proposed in what is known today, as the "Fraud Triangle" the third factor that may cause an otherwise honest person to commit fraud is the so-called "rationalization". As evidenced in many fraud related

corporate failures, for example, organizational culture became a determining factor behind the decision to commit fraud. Excuses such as “I am helping my organization”, “everyone is doing it”, “nobody gets hurt”, to name a few, are commonly used to justify such a decision. Of course, not all excuses for corporate failures are part of fraud rationalization. A competent forensic accountant

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needs to be able to identify which of them are legitimate and which are fraud rationalizations. The understanding of the moral system as well as the existing legal and regulatory framework will help a forensic accountant in identifying fraud red flags by assessing the rationalizations thereof to construct investigation hypotheses to be tested later on.

The essential attributes of forensic accountants-Better, faster, smarter: developing a blueprint for creating forensic accountants Hendi Yogi Prabowo- Journal of Money Laundering Control Vol. 16 No. 4, 2013 pp. 353-378

Knowledge requirement (KR) and task performance fraud risk assessment

Prior literature argued that the KR of the auditor is limited in terms of the nature and scope of the audit assignment. The International Education Standard (IES) No. 8, competence requirements of professional accountants sections 36-41 classified the knowledge content of the auditor into three distinct levels (IFAC, 2006a). These are historical financial information audit at a higher standard, financial accounting and reporting at a higher standard and information technology (IFAC, 2006a, 2006b). Indeed, Statement of Auditing Practice No. 30 recognizes that auditors should be conscious of the prospect of fraud presence during an audit; this was not absolutely definite and left auditors with no commitment to detect fraud (Wuerges, 2011; Albrecht and Willingham, 1993).

Literature acknowledged the fact that individuals who are educated in the use of information technology, legal, investigative, criminology, psychology and accounting will exude brilliance than others in the areas of accounting records, gathering and evaluating financial statement evidence, interviewing all parties related to an alleged fraud situation and serving as an expert witness in a fraud case (Wuerges, 2011; Hopwood et al., 2008; Singleton et al., 2006). This study also agrees with Daniel and Lee (2006) that other accountants may look at the charts, but forensic accountants dig deep into the body. Also, the SAS No. 99 discusses relevant fraud risk factors that might signal the existence of an intentional material misstatement, that is, fraud (AICPA, 2002). The risk factors Fraud risk assessment and knowledge requirement acknowledged include incentive or pressure, opportunity and attitude or rationalization. In essence, fraud risk assessment has a direct relationship to the effectiveness of auditors' fraud detection in an audit. Previous research in accounting has shown that much attention has been dedicated to fraud and fraud risk-related issues (Allen et al., 2006; Cushing et al., 1995).

Unfortunately, these studies found that auditors are poor assessors of fraud risk and, as a result, failed to detect fraud in financial statements (Knapp and Knapp, 2001; Joyce and Biddle, 1981). For this reason, it requires those in authority to consider the impact of changes in the external environment and within its activity model that may render internal

control less effective. Similarly, the O'Malley Commission recommends to the auditors to

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integrate fraud or forensic accounting procedures (substantive tests that were directed at the possibility of fraud) on every audit to improve the likelihood of discovering financial statement fraud (POB, 2000).

The importance of the acquisition of foundational forensic accounting

Skills that will enhance the likelihood of fraud detection. Others believe that adding forensic skills to the audit process assists in fulfilling the expectation gap because clients and shareholders believe these procedures are already being performed (Zikmund and O'Reilly-Allen, 2007; Coenen, 2006). This study provides evidence that stakeholders of the accounting profession understand the need for future adaptation in the new audit environment. Finally, educational leaders have stressed the need for better forensic accounting skills for auditors as a result of the implications of the changed regulatory and audit standard setting environment (Arens and Elder, 2006). The role of the auditor in the detection of fraud has a long history of controversy and confusion (Zikmund and O'Reilly-Allen, 2007). Historically, the response from the US accounting profession over the past 20 years has been the issuance of three fraud standards; SAS No. 53 (AICPA, 1988), SAS No. 82 (AICPA, 1997) which was replaced by SAS No. 99 (AICPA, 2002).

Currently, the PCAOB (2007) issued a release emphasizing that an important focus of the board is the auditor's responsibility for the detection of a material misstatement caused by fraud. Auditor adaptation to this new environment appears to be necessary to advance the audit franchise of the accounting profession.

1.6 Standards, Legal Procedure, and Professional Ethics in Forensic Accounting: Forensic Accountant need to follow.

Forensic accountant must follow certain standards to ensure integrity, competency, confidentiality, and privacy. Otherwise, the work of the forensic accounting team is at risk of becoming tainted, not persuasive, and not credible. Prior to 2002 the American Institute of Certified Public Accountants (AICPA) established Generally Accepted Auditing Standards (GAAS) for publicly traded companies. The Public Companies Accounting Oversight Board (PCAOB) was created in 2002 following the collapse of Enron and Arthur Andersen as part of the Sarbanes-Oxley Act of 2002. The PCAOB now determines GAAS. While the states regulate public accounting, neither the states nor the Federal government regulate either the forensic accounting profession or the forensic accounting industry. As a result, anyone can call him or herself a forensic accountant, fraud examiner, or any similar title, subject only to not using "public" and "accounting" in the title, and not violating the copyright or trademarks of other organizations.

The following section discusses the profession and the industry. ...accountant to be successful (Williams, 2002). Therefore, many have chosen to acquire the necessary symbolic capital by

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obtaining a degree or certificate in forensic accounting. There are approximately 155 colleges or universities in the U.S. that have degree programs in forensic accounting, majors or minors in forensic accounting, or certificates in forensic accounting (Seda, & Peterson-Kramer, 2014).

Although not a forensic accounting corporation 9, the AICPA¹⁰ issues a forensic accounting certification – Certified in Financial Forensics (CFF). (One must be a CPA to be eligible to receive the CFF since membership is restricted to licensed CPAs.). According to the AICPA, the domains for the CFF include the following: CPA Professional Responsibilities in Civil and Criminal Matters; Laws, Courts and Dispute Resolution; Reporting, Experts and Testimony; Bankruptcy, Insolvency and Reorganization; Computer Forensic Analysis; Economic Damages Calculations; Family Law; Financial Statement Misrepresentations; Fraud Prevention, Detection and Response; Valuation (AICPA, 2013c).

- Regulation of Forensic Accounting Services

- Legal services and expert services unrelated to audit (Rule 2-01(c)(4)(x) of Reg- SX of SEC: No advocacy rule

- Forensic services can be used to * extend audit procedures "shadow" audit client's Independent legal counsel and retained outside forensic accounting investigators *
- perform combinations of the above two

- Can investigate suspected illegal acts at the request of the audit committee in situations not involving litigation or regulatory proceedings

- Forensic accounting investigative services to aid management or audit committee arraying out its corporate governance responsibilities

- Forensic services already under way when government investigation commences so long as the auditor controls the work

- No services to client's legal representative in connection with investigation by SEC's Division of Enforcement, in litigation processing, or other government investigations

- Forensic accountants can not take direction from client's counsel or aid such counsel in any way

- Integrity and Objectivity

- Professional skepticism

In addition, the top ten information on Islamic values (ethics) for public administration (Ismail, 2009; Salleh, et al., 2011) which provide greater accountability, integrity and transparency in the public services are 1) justice 2) discipline 3) cooperation 4) accountability 5) obligation of duty 6) worth of character 7) consistency 8) dedication 9) integrity 10) a vigorous and mastery of knowledge.

1.7 Typical Approach to a Forensic Accounting Assignment.

- Reviewing public documents and doing Background Checks

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- Conducting interviews

- Gathering information from Trustworthy sources

- Analysing evidence

- Surveillance

- Going undercover

- Analysing the financial statements

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1.8 The Distinctive Role and Responsibilities of the Auditor and the Forensic Accountant

RULING ABOUT THE ROLE OF AUDITOR IN DETECTING FRAUD

However, the auditor's role in detecting fraud has a lengthy saga of debate. Dating back to the Lawrence Dicksee 1892 textbook "Auditing; A Practical Manual for Auditors" (the authorized American Edition was published in 1905) which lists the objective of an audit as threefold, the:

- (1) detection of fraud;
- (2) detection of technical errors; and
- (3) detection of errors in principle.

Dicksee (1905) goes on to emphasize that fraud detection is the most important component of the auditor's job. In addition, the author proclaims that the auditor who is able to find fraud is a better man than the auditor who does not find fraud (Dicksee, 1905). However, in 1895, an English Court ruled in the London and General Bank case that auditors are not expected to detect all fraud present in an audit but should perform their audits with reasonable care (Zikmund and O'Reilly-Allen, 2007). Considering for the past 100 or so years auditing practice has been operating under a reasonable care standard that has contributed to the divergence between the auditors, and a public stakeholder that misguidedly believe auditors "can, must and will find fraud" if it exists in the company they are auditing (Coenen, 2006). Similarly, 12.2 Securities Exchange Act (1934) sec 10A The following is from a PWC Newsletter:

The US Private Securities Litigation Reform Act of 1995 added Section 10A to the Securities Exchange Act of 1934. Section 10A added the auditors' responsibilities in reporting fraud to the securities laws by requiring that:

If an auditor becomes aware of a possible illegal act, the auditor must determine whether it is likely that an illegal act has occurred. If it is likely that an illegal act has occurred, the auditor must, as soon as practicable, inform the appropriate level of management and assure that the audit committee is adequately informed of such illegal act, unless the illegal act is clearly inconsequential.

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If the illegal act has a material effect on the financial statements, the auditor must, as soon as practicable, formally notify the board of directors if senior management has not taken, and the board has not caused management to take, appropriate remedial action. Section 10A also contains a "safe harbor" protecting an auditor from liability in a private action as a result of his or her reporting an illegal act to the SEC.

Nevertheless, ISA240 categorically states who is responsible for prevention and detection of fraud as follows:

- ☐ 'The primary responsibility for the prevention and detection of fraud rests with both those charged with governance of the entity and management.'
- ☐ 'An auditor conducting an audit in accordance with ISAs is responsible for obtaining reasonable assurance that the financial statements as a whole are free from material misstatement, whether caused by fraud or error'

Jeanette Van Akkeren, Sherrena Buckby and Kim MacKenzie School of Accountancy, Queensland University of Technology, Brisbane, Australia -The essential attributes of forensic accountants and to construct the forensic accountant “blueprint” as a reference for forensic accounting education and training-

Roles of Auditor and the Forensic Accounting Investigator

The auditors’ concerns include:

- Provide reasonable assurance that
- material misstatements do not exist and
- if they do exist, they are detected
- Auditors generally concern themselves with deterrence only to the extent that lack of such deterrence can entail possibility of fraud. Auditors also are generally not concerned with investigations except in case of special engagements.

The Forensic Accounting Investigators’ concerns, on the other hand, include:

- relatively narrow boundaries detailed development of factual information (from documentary as well as testimonial evidence)
- assessments of loss or damages based on the examination of the evidence
- recommendations to provide deterrence in the future

Forensic Accounting Investigator’s findings and recommendations can be used in litigations as well as testimony before federal/state

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government and judicial agencies.

Both auditors and the forensic accounting investigators need to have:

- knowledge of the industry, the company and its business practices
- knowledge of GAAP in the jurisdiction
- skills in the interpretation of business documents and records
- independence

and objectivity Some

observations:

Generally,

- however, auditor/client relationships are not adversarial, but the relationship between the forensic accounting investigator and the client are adversarial

Auditor

- is like a patrol officer; a forensic accounting investigator is like a detective. Patrol officer is always on patrol, a detective, however, is called on only when there is suspicion of fraud or fraud has already be detected.

Distinction between a Forensic Accountant and Auditor

Pat Kelley, eHow Contributor wrote on five questions on how to distinguish Between a Forensic

Accountant & Auditor Forensic accountants and auditors have similar jobs, and many forensic accountants come from an auditing background. Both are kinds of accountants. The key difference, however, is that forensic accountants are hired to investigate specific allegations of fraud. There are questions that you can ask to distinguish the two.

1. Ask Questions

Ask this question: Is there a specific allegation of fraud in a company? Forensic accountants are hired to investigate allegations of fraud. Auditors generally perform routine inquiries, generally on an annual basis. The Sarbanes-Oxley Act, passed in response to the nation's large financial scandals and Enron and WorldCom, requires company executives to certify that their financial reports are free of material misstatement and fraud. A material misstatement is an error or omission large enough to cause an investor to change an investment decision. If there is a widespread allegation of theft or misappropriation, a company may have engaged a forensic accountant.

2. What are they working on?

Examine what the auditor or forensic accountant is looking at. Auditors try to look at a representative sample of transactions, working under the assumption that if a few transactions follow generally accepted accounting principles, and then all similar transactions likely follow

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those principles. Forensic accountants, however, will look at nearly every transaction in their area of inquiry, checking to whom money was sent and whether there was adequate documentation for a transaction or series of transactions. They are also much more proactive and sceptical in their confirmation of financial transactions.

3 What do the lawyers say?

Ask who the accountant or auditor reports to. Forensic accountants usually report to a corporation's general counsel. Auditors report to a corporation's audit committee.

4. Looking at the statements

Consider the scope of services the accountant is performing. Auditors are tasked with ensuring that financial statements are free from material misstatement. To the extent they are interested in any individual transaction, they want to know whether that transaction and its treatment by a company conforms with generally accepted accounting principles and how that transaction relates to the company's financial statements. Forensic accountants, who at times can be hired to review financial statements if there are concerns about management fraud, may also target a smaller fraud

that is not material to a company's financial statement.

5. What did they study?

Ask the accountant about their training. Auditors generally have backgrounds in accounting. A forensic account has a background in accounting, criminology and law.

UNIT 2 INTRODUCTION TO CYBER FORENSICS

2.1 Overview of Computer Forensics and Cybercrime

2.2 Traditional Computer Crime: Early Hackers and Theft of Components

2.3 Contemporary Computer Crime

2.4 Conclusions and Future Issues

2.1 Overview of Computer Forensics and Cybercrime

Learning Objectives

After reading this chapter, you will be able to do the following:

- Explore the changes in society associated with the advent of technological changes and the introduction of the Internet.
- Identify the challenges associated with the enforcement and prosecution of computer crime.
- Examine the extent of computer crime in society.
- Familiarize yourself with the categorization of computer-related crime.

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I. Introduction

Cyber forensics, which is likewise known by the name of computer forensics, is a branch of digital measurable science, which identifies with confirming found in PCs and computerized stockpiling media. The goal of the cyber forensics is to look at computerized media painstakingly in a forensically solid way with the point of distinguishing, protecting, recuperating, investigating and showing realities and suppositions about the advanced data (Shrivastava, 2017). Computer forensic analysts assist in the investigation of crimes and cybersecurity incidents. In many cases, they work to recover hidden, encrypted, or deleted information related to the case. They also safeguard the integrity of data by following a chain of custody that ensures its admissibility in court. From a technical standpoint, the main goal of computer forensics is to identify, collect, preserve, and analyse data in a way that preserves the integrity of the evidence collected so it can be used effectively in a legal case. Hence, computer forensic is important because it can save organization money.

Historically, the world has experienced periods of great enlightenment and progress. The Industrial Revolution, for example, brought unprecedented knowledge and opportunities almost two centuries ago. Similarly, the automation of printing and the introduction of mass media greatly enhanced information dissemination by increasing the availability of reliable and credible sources of knowledge. Unfortunately, it also increased levels of physical lethargy, obesity, complacency, desensitization, child poverty, and criminal behavior. Today, American society has experienced similar transformations as a direct result of the Information Revolution. Consequently, a global marketplace has emerged, in which fresh ideas and increased appreciation for multiculturalism have flourished due to the introduction of the Internet which has created unparalleled opportunities for commerce, research, education, entertainment, and public discourse.

The introduction of computerized encyclopedias, international consortia, worldwide connectivity, and communications has greatly enhanced quality of life for many individuals. Indeed, the Internet can be utilized as a window to the world, allowing individuals to satiate their curiosity and develop global consciousness. It allows individuals to experience those things that they have only dreamed about. can find answers to the most complex legal or medical questions or search for their soul mates. They can download coupons for their favorite restaurants or search for recipes to their favorite dishes. In addition, individuals, corporations, public organizations, and institutions can

more effectively advertise their products or services, using graphically highlighted information and providing links to supplemental information or support. In fact, computerized access to unprecedented information has cut across traditional boundaries of communication.

Like other institutions, law enforcement has also benefited. The Internet has successfully created a nonthreatening platform for information exchange by community residents. In addition, the speed and efficiency has enabled agencies to communicate with other agencies on a global scale, solidifying relationships and increasing cooperation. Indeed, law enforcement has been able to

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further its mission by simply extending the range of audiences to whom it can communicate. Textual descriptions and graphic images of wanted suspects or missing persons can be viewed by anyone with an Internet connection, and concerned citizens can report suspicious activity in an efficient and effective manner. However, the Internet and the increasing reliance on digital technology and communications have also had negative repercussions—creating seemingly insurmountable obstacles for law enforcement. Indeed, the same technology that allows access to favorite recipes from Madagascar can be utilized to download blueprints for weapons of mass destruction. Those same individuals surfing the Web for vacation specials can stalk and harass targeted victims while enjoying the fruits of such searches. Indeed, the very advantages that make the Internet, wireless technologies, and smart phones so attractive are often the same that pose the greatest risk.

Nevertheless, an Internet comes with disadvantages that include an increasing dependence on cyber information. Many students rely exclusively on “knowledge” gleaned from Electronic sources. Unfortunately, the quality of information found in cyberspace is often questionable, and displacement of humanity has resulted from this dependence on artificial intelligence. More importantly, new technologies have a history of new forms of socially undesirable behavior, while enhancing traditional ones. Just as the automation of the printing press and the introduction of mass media exponentially increased the distribution of and demand for criminal contraband, like pornography and illegal substances, the Internet has established a virtual cornucopia of child exploitation and obscenity and has created an underworld marketplace for drugs and weapons. Moreover, the increasing availability of wireless technologies, social networking, and smart phones has complicated the investigative landscape even further, and authorities across the globe are struggling to create and enforce laws and regulations inclusive of emergent technology.

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II. Cyberspace and Criminal Behavior

Cyberspace may be defined as the indefinite place where individuals transact and communicate. It is the place between places. Telephonic conversations, occurring across time and space, were pre-dated by wire exchanges. However, the new medium known as the Internet has monumentally increased the physicality of the virtual world, outpaced only by the exponential growth in the number of users. In 2009, for example, approximately 78 percent of the United States actively used the medium as compared to 10 percent in 1995. In the UK, the growth was even more evident with

users of the medium rising from 1.9 percent in 1995 to 83.2 percent in 2009.⁵ No other method of communication converges audio, video, and data entities so effectively. Unlike traditional methods, the Internet combines mail, telephone, and mass media. As stated previously, it exposes individuals to a myriad of new ideas and may serve as a social gathering place, a library, or a place to be alone.

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Unfortunately, this virtual world is often perceived as a painless alternative to worldly problems, where individuals shed their worries and become perfect in their profiles. Privacy advocates have often overlooked the negative repercussions of this global medium, arguing zealously that the potentiality of emerging technology precludes governmental interests in monitoring citizens.

As stated previously, technological advancements have historically led to criminal innovations. Just as the Industrial Revolution enhanced threats to national security and created an environment conducive to street/predatory crime through the concentration of the urban population, the Information or Digital Revolution has created a new forum for both terrorist activity and criminal behavior. Indeed, this latest technological era has exacerbated the vulnerabilities of government institutions and personal residences alike. Critical infrastructures, increasingly characterized by tight couplings and interdependency of IT, emergency services, public utilities, banking sectors, food supplies, and transportation systems, have resulted in an interconnectivity inconsistent with traditional security strategies. In fact, it may be argued that the Digital or Information Revolution has created a criminogenic environment in which traditional criminals adapt and new criminals emerge.

III. Clarification of Terms

Just as debates rage over the appropriate codification of crime committed via electronic means, controversy surrounds the actual semantics associated with the phenomenon. For clarification purposes, then, it is necessary to define the historical usage of terms associated with technological or electronic crimes. Computer crime has been traditionally defined as any criminal act committed via computer. Computer-related crime has been defined as any criminal act in which a computer is involved, even peripherally.

Cybercrime has traditionally encompassed abuses and misuses of computer systems or computers connected to the Internet which result in direct and/or concomitant losses. Finally, digital crime, a relatively new term, includes any criminal activity which involves the unauthorized access, dissemination, manipulation, destruction, or corruption of electronically stored data. As data may be accessed or stored in a variety of ways and in a variety of locations, digital crime may be characterized as any of the three depending on case characteristics. While computer crime and computer-related crime will be used interchangeably throughout the text, cybercrime will only be used to describe that criminal activity which has been facilitated via the Internet. It would not be appropriate, for example, to categorize a residential burglary as a computer crime, even if a computer was among the items stolen. At the same time, the hijacking of an entire shipment of computer hard drives is more appropriately situated elsewhere. And, finally, the theft of millions of dollars via computer hacking is most properly denoted as a "cybercrime."

However, in all of these situations, a forensic computer scientist may be helpful. Accordingly,

this book has attempted to identify those crimes in which a computer specialist might be helpful and has used the terms computer crime and computer-related crime interchangeably.

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Additionally, students should be advised that a variety of definitions exist, and that such variations have resulted in confusion among legislators and investigators alike. Some authors, for example, argue that any crime that involves digital evidence may be characterized as a computer crime. This is misleading at best and self-serving at worst. Traditional kidnapping cases in which ransom demands are communicated via telephone will always represent a crime against a person and should not be characterized as a “telecrime.” While it is desirable to establish an environment where computers are viewed as potential evidence containers in any case, to redefine traditional predatory crime as cybercrime or computer crime is absurd. Extortion is extortion and will remain such regardless of the method employed to communicate the threat. The result of such hyper-definition is to negate some emerging legislation. This is not to suggest that legislators should cease efforts to specifically criminalize computer-specific criminal activity. Indeed, further legislation should be pursued to enhance prosecutorial toolboxes, not to replace or supplant traditional mechanisms.

Just as confusion exists regarding the appropriate terminology for crimes involving computers, the nomenclature of the science developed to investigate such activity lacks universality. For clarification purposes in this text, computer forensic science, computer forensics, and digital forensics may be defined as the methodological, scientific, and legally sound process of examining computer media and networks for the identification, extraction, authentication, examination, interpretation, preservation, and analysis of evidence.

IV. Traditional Problems Associated with Computer Crime

Although the threat of wide-scale criminal activity via computer has existed for decades, government officials tended to overlook the seriousness of phreaking and hacking prior to the mid-1980s. In fact, computer crime was all but ignored until a variety of cases exposed the vulnerability of data systems and outlined potential cataclysmic repercussions for national security. Unfortunately, such legislative and enforcement apathy created an environment conducive to criminal activity.

The first event to signal the potential of computer crime occurred in 1986, when an accounting error of less than one dollar was investigated by a dedicated employee at the University of California at Berkeley. This internal investigation revealed that a German hacker in the employ of the KGB had tapped into a military database and obtained sensitive (but not classified) information. Using only a personal computer and a basic modem, this individual was able to connect to Berkeley computers via an independent data carrier (i.e., Tymnet). Once connected, the hacker was able to move about the MILNET system with remarkable ease and relative impunity. Individuals seeking a crime have always displayed a remarkable ability to adapt to changing technologies, environments, and lifestyles. This adaptability has often placed law enforcement at a disadvantage, struggling to keep up with criminal innovations. Indeed, the law enforcement

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community has often failed to recognize the criminal potentiality of emerging technologies until it is almost too late. This trend has proven to be true in contemporary society. Fortunately, much computer-related crime involves nonspecialist users (e.g., child pornographers, narcotics traffickers, and predators). In fact, the earliest computer crimes were characterized as nontechnological. Theft of computer components and software piracy were particular favorites. Hacking, DDoS attacks, phishing, Botnets, and other technologically complicated computer crimes came later. Although the advent of technology has vastly changed the modus operandi of certain criminal elements throughout history, current advances have changed the very physical environment in which crime occurs. As such, the law enforcement community is experiencing unprecedented periods of uncertainty and ineffectiveness. Many of these problems are associated with the comprehension of the nature of the emerging technology, while others involve questions of legality and sovereignty. Unfortunately, legislative bodies and judicial authorities have been slow to respond to such inquiries, and law enforcement has been forced to develop investigative techniques without adequate legal foundations. At the same time, the lack of technological knowledge, allocated resources, and administrative apathy traditionally associated with the law enforcement community hampers even the most mundane investigation. So, while the investigators

of computer-related crime must display levels of ingenuity comparable to sophisticated criminal entrepreneurs, traditional investigators and policymakers are ill-equipped to do so. This leads to the conclusion that Although the explosion of the Internet has led to increases in global networking, information dissemination, and international commerce, its lack of physical boundaries leads to questions of sovereignty and jurisdiction.

a. Physicality and Jurisdictional Concerns

The lack of physical boundaries and the removal of traditional jurisdictional demarcations allow perpetrators to commit multinational crime with little fear (or potential) of judicial sanctions. For the first time, criminals can cross international boundaries without the use of passports or official documentation. Whereas traditional criminal activity required the physical presence of the perpetrators, cybercrime is facilitated by international connections that enable individuals to commit criminal activity in England while sitting in their offices in Alabama. In addition, electronic crime does not require an extensive array of equipment or tools. It does not require vehicular transportation, physical storage capability, or labor-intensive practices, all of which increase the potential for discovery and enforcement. In addition, this shift from a corporeal environment, where items can be seen, touched, smelled, and so on, to a virtual world, where boundaries, concrete barriers, and physical items are inconsequential, has further insulated the criminal from law enforcement. In fact, the sheer intangibility of crime scenes has all but crippled many criminal investigations.

A further concern regarding the physical intangibility of computer crime involves the traditional lack of cooperation inherent in law enforcement investigations. Issues of funding, political

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platforms, and the like have traditionally reduced communication and cooperation among jurisdictions.

b. Perceived Insignificance, Stereotypes, and Incompetence

Investigators and administrators have displayed great reluctance to pursue computer criminals. A lack of knowledge coupled with general apathy toward cybercriminality has resulted in an atmosphere of indifference. Many stereotype computer criminals as nonthreatening, socially challenged individuals (i.e., nerds or geeks) and fail to see the insidious nature of computer crime; 36.3 percent of officers believe that the investigation of computer crime interferes with their ability to concentrate on “traditional” crime.⁸

c. Prosecutorial Reluctance

Like their law enforcement counterparts, many prosecutors, particularly those in local, nonmetropolitan areas, lack sufficient knowledge and experience to effectively prosecute computer crime. Traditionally, federal and local prosecutors alike did not perceive electronic crime as serious and often granted it the lowest priority. (As many prosecutors are strongly influenced by the concerns of their constituents, they were, and continue to be, reluctant to deviate from the headline-catching cases of street crime or other violent crime.) This view was often created or exacerbated by the lack of judicial interest in these types of crime and the lack of training displayed by responding officers. Even those jurisdictions which granted electronic crime high priority were often thwarted in their efforts by a lack of cooperation in extradition requests, the victim’s reluctance to prosecute, the labor-intensive nature of case preparation, and/or the lack of resources for offender tracking.

d. Lack of Reporting

Although estimates vary, most experts agree that the vast majority of Fortune 500 companies have been electronically compromised to the tune of at least \$113 billion/ year, and 81 percent of all businesses have experienced some victimization, with 21 percent of that stemming from unauthorized access by insiders.¹² However, early studies indicated that only 17 percent of such victimizations were reported to law enforcement authorities. Underreporting on the part of businesses and corporations may be attributed to a variety of reasons, but perhaps the most common are exposure to financial losses, data breach liabilities, damage to brand, regulatory issues, and loss of consumer confidence. Contemporary society, characterized by increased reliance on paperless transactions, demands assurances that the company’s infrastructure is invulnerable and that confidential information remains inviolate. As such, organizations have a vested interest in concealing information, whose disclosure may threaten consumer/client confidence. Remember the chaos that erupted when choicePoint and the U.S. Office of Veterans

e. Lack of Resources

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Although computer intrusions have proven to be problematic within the corporate world, such institutions’ unwillingness or inability to effectively communicate with judicial authorities has led to an increase in computer crime. Unfortunately, law enforcement and corporate entities desperately need to cooperate with one another. Unlike their civil service counterparts, the business communities have the resources (both financial and legal) necessary to effectively combat

computer crimes.

f. Jurisprudential Inconsistency

Unfortunately, the Supreme Court has remained resolutely averse to deciding matters of law in the newly emerging sphere of cyberspace. They have virtually denied cert on every computer privacy case to which individuals have appealed and have refused to determine appropriate levels of Fourth

Amendment protections of individuals and computer equipment. This hesitation has become even more pronounced with the emergence of wireless communications, social networking sites, and smart phones. As such, obvious demarcations of perception, application, and enforcement of computer crime laws vary widely across the country, and a standard of behavior in one jurisdiction may supersede or even negate legal standards in another. Traditionally, trial and appellate courts evaluated the constitutionality of computer

crime statutes, searches, and investigations through the lens of the First and Fourth Amendment. Evaluating appropriate boundaries for free speech and establishing standards of reasonableness have varied across state and federal rulings, and an inconsistent patchwork of guidelines has resulted. For example, in *U.S. v. Finley*²¹, the Fifth Circuit Court of Appeals ruled that the post-arrest search of a suspect's cell phone, including text messages and call records, was constitutional as

... police officers are not constrained to search only for weapons or instruments of escape on the arrestee's person; they may also, without any additional justification, look for evidence of the arrestee's crime on his person in order to preserve it for use at trial.

While some federal and state courts have issued similar opinions, other courts have diametrically opposed the rationale of *Finley*. In *U.S. v. Park*,²² the U.S. District Court for the Northern District of California upheld a motion to suppress evidence which was retrieved from a defendant's cell phone more than an hour after his arrest. Although noting that this case was distinguishable from *Finley* in the timing of the search, the court opined that the contemporaneity of the search was insignificant compared to a more fundamental ideological principle.

V. Extent of the Problem

Many computer crimes violate both federal and state statutes. Although the federal Agencies are better equipped to deal with the complexities involved with high-technology crime, state agencies are also inundated by increasing requests from local agencies for assistance in the investigation and detection of computer crime. The lack of resources coupled with the array of criminal

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perpetrators on the Web has all but overwhelmed investigative agencies at all levels of government. Crimes committed via computer fall on a spectrum ranging from nuisance activities (spreading viruses, spamming, etc.) to computer-assisted criminal activity (stealing home addresses, maps, family information, etc.) to computer-initiated criminal activity (i.e., wire transfers, fraud). Nefarious purposes include white-collar crime, economic espionage, Organized crime, foreign intelligence, terrorism, sexual deviance, and technologically innovated traditional crime. Perpetrators range from suburban teenagers to disgruntled employees to

incarcerated felons. To make sense of this myriad of activities, motivations, and individuals, computer crimes have historically been divided into three categories:

(1) computer as a target, (2) computer as an instrument, and (3) computer as an incidental. While these categories often overlap, they have been most useful in discussions of high-technology crime.

While viruses and worms have become a daily concern for corporations, civic organizations, and individual users, in the same vein, trojan horses and other popular hacking tools are now readily available on the Web, and the theft of data has become increasingly popular. Government entities and financial institutions, in particular, have proven especially vulnerable to data theft. While much of the activity involves recreational entertainment for savvy computer users, implications for international security and wide-scale financial fraud are looming concerns. Additional criminal activities which target computers or their components include software piracy and trafficking in stolen goods. Organized crime groups have recognized the potential profit from the black market in computer chips, and various cases have involved organized Asian gangtrafficking in high-dollar computer chips.

Additional crimes that have become more accessible to the masses include counterfeiting and forgery. The introduction of high-end scanners and printers has created an atmosphere ripe for the illegal reproduction of American currency and corporate or government checks. Sophisticated graphics software, popular among virtually all computer users, enable criminals to cut and paste, rearranging and

transposing figures at will. Thus, computers can be the instrument in a variety of criminal activities.

Finally, computers can be containers or storage warehouses for crime unrelated to technology.

Drug dealers and bookies, for example, may utilize popular spreadsheet programs like Lotus or Excel to more effectively organize their records. Even burglary or homicide investigations may include evidence recovered from a computer. Indeed, as computers reach every crevice of American life, it is likely that digital evidence will be found at an increasing number of crime scenes, unrelated to computer crime. Thus, it is essential that all investigators, not just those involved in high-technology units, recognize

the elevated possibility of computers as evidence receptacles.

VI. The Emergence of e-Cash: A New Problem for Law Enforcement

The past decade had witnessed an increasing proliferation of innovative payment mechanisms to facilitate e-commerce. Such innovations have utilized Internet and wireless devices, and the

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migration from paper to electronic payments has reached all corners of the globe. Like all other emerging technologies, the implications for e-payments are both positive and negative. Consumers have benefited, for example, from the enhanced services and efficiency offered from e-banking. In addition, the low overhead associated with online financial institutions has increased competition, resulting in lower interest rates and higher yields. On the other hand, the criminal element has embraced a variety of new payment methods which are often anonymous, involve multijurisdictional transactions, and exist in an environment which lacks regulation and government oversight. These characteristics facilitate money laundering and terrorist financing,

and make it extremely difficult for investigators attempting to “follow the money.” According to the Financial Action Task Force (FATF), these new payment methods include, but are not limited to, prepaid cards, electronic purses, mobile payments, Internet payment services, and digital precious metals.

- a. Prepaid Cards
- b. Stored Value Cards
- c. Mobile Payments
- d. Internet Payment Services
- e. Digital Precious Metals

Prepaid Cards

They are similar to debit cards in that they are attached to an account and provide access to monetary funds that are paid in advance by the cardholder. There are two primary types of prepaid card systems: limited or closed and multipurpose or open. Limited purpose or closed system cards may be used only for a finite number of purposes, and are issued by a particular merchant, telecommunications provider, or transit company. Multipurpose or open system cards, on the other hand, may be used for a wide range of purposes, may cross geographic boundaries, and may be used by any user. They are typically associated with a card payment network, like Visa or MasterCard, which may be attached to a particular depository account or linked to a line of credit by another merchant.

Stored Value Cards

Also known as electronic purses, these are cards whose value is stored electronically on the device via an integrated circuit chip. Unlike magnetic strips, which only store account information, an e-purse actually stores funds on the chip. In fact, the user is literally carrying her funds with her, just as she does when she places money in her purse. This method of payment is extremely convenient, as online connection and cardholder identification are not necessary as the transaction vehicle was designed to substitute for cash. Such purses are generally reserved for micropayments, such as those used for public transportation, parking tickets, or vending machines.

Mobile Payments

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Mobile payments are payments made via mobile phones or other wireless communication devices. Some of these transactions are activated using voice access, text messaging protocols, or wireless application protocols.³⁶ This payment method is more widespread in Southeast Asia and Europe, although its use is becoming more popular in the United States. A second type of mobile payment involves the use of a telecom operator as a payment intermediary which authorizes, clears, and settles the payment. Most typically, these transactions occur when the telecom operator authorizes the consumer to charge the transactions to the phone bill. Internet Payment Services There are two types of Internet payment services which are becoming increasingly popular in the United States. The first of these involve payments which rely on a bank account and use the Internet as a means of transferring funds to or from an established financial account. The second are those payments which are

provided by nonbank institutions

operating exclusively on the Internet that are only indirectly associated with a bank account. One of

the most popular, PayPal, serves as an intermediary for individuals and organizations that wish to effect transactions via the Internet. Such intermediaries establish prepaid accounts funded from credit/debit cards or credit transfers. Methods of fund disbursement vary among these intermediaries, with some issuing checks and others issuing credits.

Digital Precious Metals

They represent one of the newest forms of commodities trafficked on the Internet. It involves the exchange of options or the right to purchase a designated amount of precious metals at a particular price. The rationale for this type of currency involves the avoidance of currency fluctuation and foreign exchange. It is particularly attractive to cybercrimes for a variety of reasons. First, they differ from other online payment systems in that they are irreversible. Unlike sites like PayPal, there is no complaint or fraud area where users can recover losses due to fraudulent exchanges or user mistakes. Second, some have refused to adhere to applicable banking laws. Finally, many virtual dealers allow users to maintain anonymous accounts. This fact, coupled with the dichotomous nature of the process (i.e., the first level are the digital precious metal dealers and the second level involves a digital precious metal exchange service), exponentially increases the difficulties of criminal investigations and provides an attractive venue for money launderers. However, that does not suggest that prosecution is not possible. E-gold Ltd., a digital gold currency operator, was prosecuted by the U.S. Department of Justice for moneylaundering in 2008. For all intents and purposes, the company is currently defunct. Unfortunately for legitimate users, they have been unable to access the value in their account since the company's assets were frozen during the investigation. It is likely that other companies will emerge to take their place.

2.2 Hierarchy of Contemporary Cybercriminals

Although the evolution of computers and global communications has dramatically broadened the population of cybercriminals, the trend in categorizing them by their level of sophistication and/or

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their motivation has not. There are five general categories of cybercriminals in today's society: script kiddies, cyberpunks, hackers/crackers, cybercriminal organizations, and hactivists. (While some of the categories are primarily based on sophistication and others on motivation, the categories are not necessarily mutually exclusive.)

Script kiddies, also known as skidiots, skiddie, or Victor Skill Deficiency (VSD), are the lowest life form of cybercriminal. The term is a derogatory one used by more sophisticated computer users to refer to inexperienced hackers who employ scripts or other programs authored by others to exploit security vulnerabilities or otherwise compromise computer systems. Technologically the least sophisticated of all cybercriminals, script kiddies are generally not capable of writing their own programs and do not fully understand the programs which they are executing. Thus, they are not capable of targeting a specific system, but are limited to those targets which possess the identified vulnerabilities. The least advanced of this category even employ prepackaged software

like Deep Throat or DeepBus. Motivations for script kiddies can range from simple pranks, as when college students use Trojans to remotely “hide” their friends’ term papers, to criminal profit, as when users capture bank account and password information to access a victim’s account. Cyberpunks is an innocuous term which has been hotly contested by First Amendment advocates but has been used by law enforcement officials to refer to individuals’ intent on wreaking havoc via the Internet. The term was initially used to refer to an emerging genre which marries science fiction, information technology, and radical change in the social order. However, law enforcement authorities often use it as a category which includes vandalism, destructive programs (e.g., viruses and worms), and general mischief for no economic gain. Sophisticated computer criminals who are capable of programming, writing code, and breaching complex systems are categorized as hackers or crackers depending on their motivation. Hackers, as previously discussed, are those individuals who identify and exploit system vulnerabilities but who lack economic motivation. Crackers, on the other hand, are those sophisticated users who employ their knowledge for personal gain. Originally known as criminal hackers, the term derived from a combination of the two terms.

Cybercriminal organizations are those groups comprised of criminally minded Individuals who have used the Internet to communicate, collaborate, and facilitate cybercrime. Their motivations are never innocuous and include those activities associated with political extremism or economic gain. The sophistication of the methods employed and the technical expertise of their members range from elementary to highly complex. The term does not include traditional organized crime syndicates.

Criminal hackers, or crackers, are those who target data which is valuable on its face (e.g., trade secrets and proprietary data) or directed at data (e.g., credit card data) which may be used to further

other criminal activity. Unfortunately, many users recognize the potentiality for exploitation of valuable data and include at least a modicum of security, but fail to appreciate risks associated with other forms of data. In fact, the data targeted may appear to be totally benign or innocuous.

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In reality, the level of intrusion and the nature of the objective may pose risks ranging from physical security to operations security.

The final category of hackers is also the fastest growing. In fact, hacktivists accounted for the majority of all compromised records in 2011. The term hacktivism emerged in the 1990s when the Cult of the Dead Cow hacker collective coined the term to describe their actions. In contemporary parlance, the term is used to describe technological social movements. According to Verizon’s 2012 Data Breach Investigations Report,⁷ The major shift that occurred in 2011 was that activist groups added data breaches to their repertoire with much heightened intensity and publicity. In other words, 2011 saw a merger between those classic misdeeds and a new “oh by the way, we’re gonna steal all your data too” twist.

Defining Hacking

The root of the term hacking has been claimed by the Massachusetts Institute of Technology (MIT), and dates back to the 1960s when the term was used by MIT students to refer to either the

development of novel techniques to identify computer shortcuts or clever pranks. (Early examples of such pranks were not always computer related, and included placing a full-sized model of a patrol car atop the dome on Building 10.) Most probably, the term derived from the metaphor of hacking away at an object until it

gave way. In the competitive culture of the MIT campus, innovative computer solutions inevitably led to even more sophisticated techniques as students attempted to outdo one another. In the 1980s,

the term was popularized in the film *War Games*, and the hacker subculture exploded.

Hacking is a global phenomenon and is not restricted to the United States. Hackers have been found in virtually every country in which computer technology is available.

Remarkably, these individuals, irrespective of national origin, display startling similarities.

Evolution in the Hacking Community

The emergence of the term in the popular media coupled with the increase in accessibility and connectivity removed the ivory towers and dramatically increased the number of individuals engaged in hacking activity. The newcomers to the fray were often young, socially retarded males who initially became enamored of computers and computer technology through role-playing games. Such entertainment, necessitating excessive downloads, led these individuals to manipulate

telephone exchanges. (Prior to the mass introduction of Internet service providers and unlimited access, calls via modem tended to be quite costly.) Common justifications for their actions included an antiestablishment ideology that inferred that corporate structures and government entities were designed to abridge individualism and discourage collective unity. (Thus, hacking organizations may be likened to Christian identity groups which believe that a government conspiracy to exploit American citizens is furthered by corporate assistance.) In addition, early hackers emphasized the virtuality of cyberspace, arguing that the Internet is a sphere of unreality,

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where nothing is concrete and everything is simulated. Finally, traditional hackers around the globe shared a sense of overwhelming empowerment in which they were the keepers of all knowledge. Thus, the traditional hacker culture was characterized by an antiestablishment rhetoric—a feeling that was largely shared by a generation. Although the verbiage and sophistication of articulation varied, a statement made by Lloyd Blankenship (aka, the Mentor) encapsulated the angst felt by the subculture.

This is our world now ... the world of the electron and the switch, the beauty of the baud. We make use of a service already existing without paying for what could be dirt-cheap if it wasn't run by profiteering gluttons, and you call us criminals. We explore ... and you call us criminals. We seek after knowledge ... and you call us criminals. We exist without skin color, without nationality, without religious bias ... and you call us criminals. You build atomic bombs, you wage wars, you murder, cheat, and lie to us and try to make us believe it's for our own good, yet we're the criminals. Yes, I am a criminal. My crime is that of curiosity. My crime is that of judging people by what they say and think, not what they look like. My crime is that of outsmarting you, something that you will never forgive me for. I am a hacker, and this is my manifesto. You may stop this

individual, but you can't stop us all.

2.4 Conclusions

Just as the introduction of the telephone gave American society the first wave of heavy breathers, telemarketers, and rapid response, the creation of the Internet has resulted in a myriad of developments, some positive, others negative. Individual users can travel all over the world at the touch of a button. They can access the latest sports scores, stock prices, and international news, while downloading their favorite music or photographs. The proliferation of technology and the increasing reliance and interconnectivity which characterize contemporary society has resulted in a criminal landscape where the existence of computer crime and digital evidence is increasingly the norm. White-collar criminals have increasingly employed technology to facilitate their embezzlement and bank fraud schemes. Entrepreneurial organized crime groups have utilized the Internet and cell phones to communicate and further the dissemination of criminal contraband. They have also used technology to replace archaic bookmaking and recordkeeping methods with spreadsheets and financial software. Even street criminals have embraced modern innovations trafficking in stolen computer components, fencing stolen goods in cyberspace, and using the medium to identify and target potential victims. The explosion of smart phones, the increase in wireless communications, and social networking addictions have only exacerbated the problems. In fact, it may be argued that the perceived anonymity of the Internet coupled with the accessibility to technology has resulted in an environment far more criminogenic than previous eras. However, law enforcement authorities are struggling with outdated technology, a lack of significant resources, and administrative and public apathy.

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UNIT 3 FRAUD, FRAUDSTER AND THE FRAUD VICTIMS

UNIT 3: FRAUD, FRAUDSTER AND THE FRAUD VICTIMS

The fraud problems

Fraud has been in existence for generations. For example, the so-called "financial scandals" have plagued the world's economy since before the Industrial Revolution (Pearson and Singleton, 2008). Many of the scandals (such as that of the equity funding in 1970 where computers were used to as a means in perpetrating fraud) became milestones with historical importance in our journey in combating fraud (Pearson and Singleton, 2008). According to the Black's Law Dictionary (2009), fraud is a knowing misrepresentation of the truth or concealment of a material fact to induce another to act to his or her detriment [. . .] A misrepresentation made recklessly without belief in its truth to induce another person to act. As argued by Ramamoorti (2008, p. 522): Fraud involves intentional acts and is perpetrated by human beings using deception, trickery, and cunning that can be broadly classified as comprising two types of misrepresentation: *suggestio falsi* (suggestion of falsehood) or *suppressio veri* (suppression of truth).

The essential part of any fraud is the use of deception to obtain benefits. As defined by Wells (2005, p. 8), four general elements must be present for an offence to be called "fraud":

- (1) a material false statement;
- (2) knowledge that the statement was false when it was uttered;

- (3) reliance on the false statement by the victim; and
- (4) damages resulting from the victim's reliance on the false statement.

There are many types of fraud in the society and no single classification system in the world can comprehensively identify all of them as far as literatures are concerned. The Association of Certified Fraud Examiners (ACFE), for example, has been using the Fraud Triangle system in its biannual global studies on fraud (Association of Certified Fraud Examiners, 2012b, a). The system divides fraud into three major categories: corruption; asset misappropriation; and financial statement fraud (Association of Certified Fraud Examiners, 2012b). From the existing cases, there are many ways fraud can be perpetrated, from the simple misuses of trust to the sophisticated computer based offence. In addition, by nature, fraud is an interdisciplinary issue requiring the understanding of a number of different areas of knowledge for prevention and investigation. Whereas the Fraud Triangle explains about the "why" element of fraud, the Fraud Element Triangle focuses more on "how" it is perpetrated.

Types of Fraud

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Fraud is a complex and dynamic problem. Every nation has its own specific fraud problem. In less-developed countries like Nigeria and Indonesia to mention but a few, corruption is known to be the most common or at least the most exposed type of fraud. The Indonesian Corruption Eradication Commission (KPK) in its fraud statistics classifies corruption into five categories: procurement of goods and services; bribery; budget misappropriation, unauthorized collection and inappropriate licensing (Corruption Eradication Commission, 2012). Among these categories, procurement of goods and services is known to be the most prevalent offence. One thing worthy of a note is that despite the high level of corruption in Indonesia, it is only the tip of a huge iceberg. This is so since media coverage is focusing primarily on corruption in public sectors whereas, in reality, it is also a pandemic in the private sectors. In public sectors, it seems that new corruption cases emerge everyday perpetrated by high profile public figures such as members of the parliament, political figures, ministers, etc.

Elements of a fraud

Essential elements of a fraud are act; concealment; and conversion (Albrecht et al., 2012, p. 81). The "act" is where offenders stole the assets, "concealment" is concerning offender's efforts to hide fraud from others, and "conversion" is when offenders spends or convert the proceeds of their crime (Albrecht et al., 2012, p. 42). Opportunity for committing fraud may come from one's position in his or her organization that can be misused for obtaining unlawful benefits (Dorminey et al., 2012, p. 565). According to the ACFE's biannual study, generally, the higher a fraud perpetrator's position in his or her organization, the more damaging his or her fraud will be (Association of Certified Fraud Examiners, 2012b).

Combating fraud

Understanding the problems In order for us to be able to formulate the most effective solution to combat fraud in the society, we first need to gain sufficient knowledge about it. Although Enron and WorldCom have always been used as references when discussing fraud and forensic

accounting, they are only representing part of the problem. To be able to solve the growing fraud problem, one needs to comprehend the nature and dynamics of fraud. Many literatures have attempted to define fraud in the way that is best representing the actual structure and dynamics of the offence. In practice, the definition of fraud is often determined by the existing legal system which, in many cases, differs across countries. For example, according to Section 380(1) of the Criminal Code of Canada provides the general definition for fraud in Canada is Everyone who, by deceit, falsehood or other fraudulent means, whether or not it is a false pretence within the meaning of this Act, defrauds the public or any person, whether ascertained or not, of any property, money or valuable security or any service:

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(a) is guilty of an indictable offence and liable to a term of imprisonment not exceeding fourteen years, where the subject-matter of the offence is a testamentary instrument or the value of the subject-matter of the offence exceeds 5,000 dollars; or

(b) is guilty:

(i) of an indictable offence and is liable to imprisonment for a term not exceeding two years; or

(ii) of an offence punishable on summary conviction. where the value of the subject-matter of the offence does not exceed 5,000 dollars.

In Indonesia, fraud has often been narrowly defined as synonymous to corruption. According to the Law No. 31 Year 1999, only offences, which cause losses to the state, are considered as corruption. As defined by the Law No 31 Year 1999 (Article 2) (The Republic of Indonesia, 1999): Anyone who illegally commits an act to enrich oneself or another person or a corporation, thereby creating losses to the state finance or state economy, is sentenced to life imprisonment or minimum

imprisonment of 4 (four) years and to a maximum of 20 (twenty) years, and fined to a minimum of Rp200,000,000 – (two hundred million Rupiah) and to a maximum of Rp1,000,000,000 – (one billion Rupiah).

This implies that the efforts to combat fraud in Indonesia are mainly focused on the corruption perpetrated by public officials. In reality, fraud comes in various sizes and shapes. The ACFE, for example, had formulated a classification system known as the Fraud Tree. In this classification system fraud is broadly categorized into corruption, asset misappropriation and financial statement fraud (Association of Certified Fraud Examiners, 2012b, p. 7). The Fraud Tree itself has been used by the ACFE in its biannual studies over the past few years. Such classification is an important first step in combating fraud as understanding a crime is a prerequisite to understanding about how to mitigate it.

Prevention, detection, and investigation and prosecution

In general, solving the fraud problems requires decision makers to consider a number of major elements: prevention; detection; investigation; and prosecution. Prevention should be designed so as to reduce the possibility of fraud to occur. A common approach is by reducing the opportunity of potential fraud offender to perpetrate their offences, which is also known as the “situational crime prevention” method (Sutton et al., 2008; Hayward, 2007). Detection involves recognizing the symptoms of fraud from the existing “red flags”. Such red flags can be in the forms of

accounting anomalies, weak internal control, operational anomalies, extravagant lifestyle and behavioral symptoms (Albrecht et al., 2012, pp. 137-155). According to ACFE's latest biannual global study on fraud, "tip" is the most common way a fraud is detected in the first place. This signifies the importance of, among other things, a whistle blowing system in an organization as a means for reporting fraud (Association of Certified Fraud Examiners, 2012b, p. 14).

Despite an organization's efforts to prevent fraud, in many cases fraud perpetrators still manage to commit their crime, which calls for fraud investigation. Although financial auditing services may

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have been used in detecting material misstatements, ordinary auditing skills are often deemed insufficient when dealing with fraud related issues. This is where the so-called "forensic accounting" plays its role.

Making Mistakes, Errors in Judgment, and Intentional Misdirection

MJ Blehart 2019

Everyone makes mistakes, which differ from errors in judgment and intentional misdirection. Congrats on being human. It is what you do after making a mistake that speaks volumes. As a part of human nature, you are going to make mistakes. Period. Nothing to be done for it, it is going to happen. You are perfectly imperfect, so mistakes are inevitable. Once you acknowledge this, it becomes a lot easier to figure out what to do after making a mistake. Of course, mistakes can come in many shapes and sizes. Some are relatively small and nearly insignificant. On the other hand, some can be pretty colossal, and impact not only you, but people you care about, and even complete strangers. It is important to recognize the difference between a mistake, an error in judgment, and intentional misdirection.

What constitutes a mistake?

To put it as simply as I can, a mistake is an error. It might be a choice, such as which route to take to work. Maybe you messed up your math and gave too much change to someone. You might have called someone by the wrong name or title. Mistakes are unintentional, and that's the most important thing to note about them. A mistake is an error of some sort that may or may not have been the result of choice.

According to Dictionary.com, a mistake is: noun an error in action, calculation, opinion, or judgment caused by poor reasoning, carelessness, insufficient knowledge, etc.

a misunderstanding or misconception.

verb (used with object)

to regard or identify wrongly as something or someone else

to understand, interpret, or evaluate wrongly; misunderstand; misinterpret.

verb (used without object)

to be in error.

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Yes, this does place the onus of a mistake on you and me. However, even when a mistake is the result of carelessness, it is an unintentional act. Everyone makes mistakes along the way because

nobody knows everything. Further, if you only took actions based on absolute knowledge, you would stifle growth potential. Mistakes can help you to learn, evolve, and discover new possibilities, realities, and so on.

How does an error in judgment differ from a mistake?

A mistake is unintentional. You did not set out to err, but it happened. No intent was behind it. An error in judgment, though, might be intentional, or might not. Let us say that you believe that a person is lazy. Plenty of evidence leads you to this conclusion, so you judge them (silently or otherwise) as such. Then you learn they are suffering from medical conditions that leave them with zero energy, and just getting out of bed in the morning is a triumph most days. Not lazy, ill. That is an error in judgment. You were mistaken, but you intentionally believed in that person's laziness. This can also apply to the self. I used to think I was an overweight, utterly unskilled and un-athletic lump of flesh, incapable of any prowess in any sport. Twenty-seven years of fencing, the resulting reaction time in other applications and not entirely sucking at fencing has proven that wrong.

An error in judgment is not always intentional, however. If you have limited information from which you form an opinion and subsequent judgment that turns out to be wrong, no intent was behind that. Unless you retain an opinion you know to be false, wrong, or hurtful, in which case you have provided it with intent.

What is an intentional misdirection?

While a mistake is unintentional, and an error in judgment can be both with and without intent, an intentional misdirection is all about intent. Simply put, this is a lie. A half-truth. A tall tale. Bullshit. Cheating on your taxes. Having an affair. Stealing. All of these are referred to as "mistakes," because they are, technically (see the Dictionary.com definition of mistake above). But the intentional action of misdirection is more than just an error, it's an error with malice of forethought. Sure, you know it is a mistake to cheat on your golf game and call a mulligan on every hole — or two. But you do it anyhow to impact your handicap and lower your score. If you get caught, you can make some excuse for your "mistake" in scoring. Big or small, intentional misdirection is not a harmless mistake. Read many of the Tweets from Trump if you want to see this in action.

Being accountable when you make a mistake or an error in judgment, and learn about it, you have a choice. Let it go without saying anything, or apologize and/or make adjustments/corrections. It is best to be accountable for your mistakes. That way, you can take corrective or alternative actions,

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and let people know that you recognize them. Further, being accountable for a mistake can help you to learn not to repeat it, and even prevent similar errors in the future. Accountability and responsibility go a long ways towards being more complete and gaining the things, you desire to have for your life. When you are not accountable for yourself and your thoughts, feelings, or actions, you give away all of your empowerment.

Why does that matter? Because if you give it away, you lessen your potential to be the best you that you can be. But more than that, you disempower yourself and set yourself up to hold some pretty negative opinions of yourself. For example, "I am a screw-up," and "I never do anything

right” and “I’m bad at everything” and even “my life is worthless because all I do is make mistakes and hurt people with them.” Blame, even of yourself, is not accountability, because it leaves no room for improvement or change. Any of this look at all familiar? Accountability looks like this: I screwed up. What can I learn from that? I did that wrong, what is the right way to do it? I may be bad at this, but I can learn and get better. My life is worthwhile and like everyone, I make mistakes and apologize if I hurt someone. Blame does not feel good, but accountability does. Ownership of your mistakes and errors in judgment opens you up to changing and doing better next time.

Finally, it is important that after you make a mistake or an error in judgment you forgive yourself. This is usually the hardest part. If you are anything like me, you may have a skewed perspective of your personal right/wrong filter. As such, when you make a mistake or an error in judgment you get upset with yourself and find it hard to be forgiving. But it is as important as being accountable to be forgiving to yourself. Again, you are only human, and mistakes are a part of your nature. Accept that part of you, acknowledge it, and forgive yourself when it happens. Yes, this can also be applied to an intentional misdirection, too. However, first you have to own up to your bullshit, be accountable for your intentional action, and make a real effort to change that action. If you continue to act with bad intention and tell lies and commit acts you know to be wrong, you probably don’t even see a need to forgive yourself in the first place. But if you “take ownership of your ugly” as Jen Sincero says, you can also forgive yourself in time.

You are perfectly imperfect. Mistakes happen, as do errors in judgment and intentional misdirection. Recognize when this happens, acknowledge it, be accountable for it, and forgive yourself along the way.

This is all a part of the process of life. The good, the bad, the ugly. The choices for what do with it and decisions on how to alter it — or not — are entirely up to you. Congrats on being human. It is what you choose to do after making a mistake that speaks volumes to everyone you encounter as well as to you yourself. You are worthy and deserving of using your mindfulness to find and/or create the reality in which you desire to live. When all is said and done you matter, as does how you handle mistakes you make.

According to MJ Blehart there Five Easy Steps to Change the World for the Better:

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1. Life Lessons
 2. Mistakes
 3. Mindfulness
 4. Self-awareness
 5. Personal Development
- (<http://www.mjblehart.com>)

D. CASE INVESTIGATION PROCEDURE

ds and the setting of goals.]

E. EVIDENCE GATHERING

-Electronic Evidence.

fraud detected during the investigation process.

concept of proof and the principles of using evidence to prove a case.

EVIDENCE

Qualities of expert evidence

The courts have variously commented on the qualities required of expert evidence. The most important of these is that the evidence be unbiased and objective. In addition, relevance, reliability and cost effectiveness of expert evidence are also important considerations.

Unbiased, objective

Expert evidence should be objective and fair. Some experts will be inclined to tailor their evidence to their client's requirements and pay insufficient attention to the need to be balanced and objective. The trial judge, if not exposed by the opposing expert's challenge to what they are saying, will usually perceive this lack of objectivity, readily in any event. In *National Justice Companion Naviera S.A. v. Prudential Assurance Co. Ltd* Cresswell J. stated:

"An expert witness should provide independent assistance to the Court by way of objective unbiased opinion in relation to matters within his expertise ... An expert witness in the High Court should never assume the role of an advocate."

In *Whitehouse v. Jordan* Lord Wilberforce said:

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"Expert evidence presented to the court should be, and should be seen to be, the independent product of the expert uninfluenced as to form or content by the exigencies of litigation." The role of the expert is to educate and inform decision-makers so that the truth is revealed. It can be argued

that the process of obtaining written reports, and examination and cross-examination of opposing experts is more likely to reveal the truth than evidence from a single 'neutral' expert. In either case, it is essential that expert evidence be given with independence and impartiality.

It can be argued that bias is self-policing, since an overtly biased expert will soon lose credibility within the professional community and not be called by other parties in future cases. But such an argument may be fragile since the reality is that parties may "shop around" for an expert who will best support their case. Furthermore, the expert's evidence may risk being coloured by a preexisting or continuing relationship that may exist between the expert and the lawyer or the litigant. Nevertheless, an expert is probably shortening his useful life significantly if he compromises his independence.

Relevant

The significance of relevance can be seen from the following extract from *R v. Wilson*:

"... lack of relevance can be used to exclude evidence not because it has absolutely no bearing upon the likelihood or unlikelihood of a fact in issue but because the connection is considered to be too remote. Once it is regarded as a matter of degree, competing policy considerations can be taken into account. These include the desirability of shortening trials, avoiding emotive distractions of marginal significance, protecting the reputations of those not represented before the Courts and respecting the feelings of a deceased's family. None of these matters would be

determinative if the evidence in question were of significant probative value.”

In *R. v. Mohan* the Court ruled that, *prima facie*, expert evidence was “relevant” if it was “...so related to a fact in issue that it tends to establish it.” A more academic definition of relevance is provided by Stephen (1946, p. 4) who says that any two facts are relevant to each other if they are: “...so related to each other that according to the common course of events one either taken by itself or in connection with other facts proves or renders probable the past, present, or future existence or non-existence of the other.”

Reliable

Expert evidence should be able to withstand close scrutiny to determine whether it is “reliable”. “[E]xpert evidence which advances a novel scientific theory or technique is subjected to special scrutiny to determine whether it meets a basic threshold of reliability and whether it is essential in the sense that the trier of fact will be unable to come to a satisfactory conclusion without the assistance of the expert.”

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Cost effective

In the same case, the issue of cost-effectiveness of expert evidence was dealt with as follows: “Evidence that is otherwise logically relevant may be excluded ... if it involves an inordinate amount of time which is not commensurate with its value or if it is misleading in the sense that its effect on the trier of fact, particularly a jury, is out of proportion to its reliability.”

DIGITAL EVIDENCE

Introduction

Digital devices are everywhere in today’s world, helping people communicate locally and globally with ease. Most people immediately think of computers, cell phones and internet as the only source of digital evidence, but any pieces of technology that process information can be used in a criminal way. For example, hand-held games can carry encoded message between criminals even newer household appliances, such as a refrigerator with a build-in TV, could be used to store, view and shares illegal images. The important thing to know is that responders need to be able to recognized and properly seize potential digital evidence.

Digital evidence is defined as information and data of value to an investigation that is stored on, received or transmitted by electronic devices. This evidence can acquired when electronic devices are seized and secured for examination.

Digital evidence:

- Is latent (hidden), like figure print or DNA evidence
- Crosses jurisdictional borders quickly and easily
- Can be altered, damaged or destroyed with little effort
- Can be time sensitive

There are many sources of digital evidence, but for the purposes of this publication, the topic is divided into three 3 major forensic categories of devices where evidence can be found: Internet-based, stand-alone computer or devices, and mobile devices. These

areas tend to have difference evidence-gathering process, tools and concerns, and different types of crimes tend to learned themselves to one devices or the other.

F. EVALUATING FINANCIAL AND NON-FINANCIAL EVIDENCE

-financial capacities.

H. CASE STUDIES IN FORENSIC ACCOUNTING

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LEARNING OBJECTIVES OF THE EXERCISE

We believe this exercise contributes to a student's development in three key areas:

- 1) Students will gain a better understanding of fraud prevention and detection.
- 2) Students will gain a better understanding of the fraud triangle (PCAOB #12): incentive, opportunity, and rationalization.
- 3) Students will develop a more thorough understanding of internal controls and how they can be used to prevent and detect fraudulent behavior.

In Auditing Standard No. 12 (2010), the PCAOB identifies three fraud risk factors (also known as the fraud triangle⁴) that auditors should consider during their risk assessment procedures because these three conditions are often present when fraud occurs. First is an incentive or a pressure to commit fraud. Second, an opportunity to perpetrate fraud. Third, an attitude or rationalization for committing fraud. For years, auditing courses have used the fraud triangle framework to introduce accounting students (and future auditors) to the concepts of fraud prevention and detection. The fraud triangle is easy to understand and simple to use; it does not require significant technical accounting knowledge to implement. Once a student understands the fraud triangle, it is intuitive and therefore memorable. For these reasons, we believe the fraud triangle is a good framework to use for this exercise and a good tool for introductory accounting students to learn and use.

Researchers have also evaluated the validity of the fraud triangle framework empirically. Hogan, Rezaee, Riley, and Velury (2008) summarize academic research findings relevant to the PCAOB's financial statement fraud project. During their analysis, Hogan et al. (2008) find that there exists a significant body of literature identifying the characteristics of fraud firms and that this literature supports the three specific components of the fraud triangle. Hogan et al. (2008, p. 246) also conclude that the three specific components of the fraud triangle are "grounded in empirical evidence." Therefore, we believe that the fraud triangle framework is useful for this assignment given the targeted participants and is also valid according to empirical evidence.

IMPLEMENTATION GUIDANCE

As previously stated, we designed this exercise to expose students in the introductory accounting course to several aspects of fraud prevention and detection using an actual fraud perpetrated at a not-for-profit organization.

We conducted the exercise in the second month of the introductory accounting course. At this point, students were familiar with the basics of financial statement preparation and analysis, but had not encountered internal controls or fraud (although some classes may

have been exposed to limited in-class comments related to fraud, it is not a part of the core curriculum). We conducted the exercise during a recitation period, which allowed

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sufficient time for both instruction and discussion. We did not ask students to prepare for the recitation and did not provide any fraud-related training in the course before this exercise. In order to be consistent across test samples, we limited the time allowed for the assignment to one hour. We found the one hour time constraint worked well, although some classes would have discussed the assignment longer if provided additional time.

We organized students into groups of three to four to facilitate discussion and outlined the intended progression through each of the phases. Next, we distributed the first phase of the exercise (Box 1 “Phase I: Background”) to the students. We allowed the students a few minutes to read through the document and then asked them to discuss within their groups the questions related to that phase of the exercise (Table 1). After approximately five to ten minutes, we began an instructor-led discussion of each question. We first called on individual groups to respond and then allowed any class member to provide input related to the question.

CASE STUDY NO 1

Box 1. NUASA Homeless Shelter Assignment (Phase I: Background).

On Friday, April 15, 2011, Ahmad Yaro the Director of NUASA Homeless Shelter, opened the purchasing card bill⁹ and noticed that the bill had a higher balance than she expected. After reviewing the detailed statement, she noticed some unexpected purchases from K-Mart. She also saw that the purchases were charged to the card assigned to Dean Brown, the administrative assistant at NUASA. Ahmad and Dean had separate purchasing cards for the NUASA account and each month two separate bills were received from the bank in one envelope, so it was easy to know what charges were made to each card. Ahmad and Dean used the cards to purchase office supplies,

coffee, tea, sugar, towels, sheets, blankets, and other small items the shelter needed. However, Ahmad did not remember sending Dean to K-Mart during the last several months. Usually Ahmad reviewed the bills each month when she signed the cheques for payment but occasionally she would sign the Cheques without reviewing the statement. With a cold feeling in her stomach, Ahmad went to the accounts payable file and pulled out the other purchase card statements. Sitting

down behind the desk in her office, Ahmad started looking carefully at the purchase card statements from the last few months. She was relieved to see nothing else out of order in the statements.

Ahmad sat back and took a deep breath. NUASA was a small homeless shelter with about 15 full- and part-time employees that took care of approximately 50 individuals each night. The shelter operated as a 501(c)(3)¹⁰ organization and Ahmad had been the Director for the past eight years. She loved her job and she loved “her family”: the employees and “residents” of the shelter. After a few minutes, Ahmad realized that she still had a problem with Dean for charges she found on

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the statement she opened earlier this morning, but he would be out of the center until the following Tuesday, so she had some time to think about what to say to him.

Ahmad relied on Dean to take care of the administration of the center. He had worked at NUASA for 11 years. He sat at the front desk each day, greeting the “residents” as they came and went. He opened the mail, collected contributions, and prepared the bank deposit slips so Ahmad could make

the bank deposits after work on Fridays. He kept the checkbook, although every check required Ahmad’s signature. He recorded all of the financial transactions to the small-business general ledger system the shelter used, and prepared the monthly reports for Ahmad and the Shelter’s Board of Directors. If he was not trustworthy ... Ahmad did not even want to think about it. But she had to think about it. Ahmad got her key, unlocked the bottom drawer of Dean’s desk, and removed the checkbook. Ahmad set the binder containing the Cheques for the shelter on Dean’s desk and flipped it open.

SECOND CASE STUDY

Box 2. NUASA Homeless Shelter Assignment (Phase II: Discovery).

Ahmad slowly turned the pages of the checkbook, carefully looking at the check stubs with Dean’s handwriting on them. The shelter only used about 15 Cheques each month for regular expenses, so she was surprised to see more than 10 check stubs marked “void” just in the last few months. She made a list of the check numbers for the Cheques marked “void” and went to the file cabinet where the bank statements were filed. Ahmad took the folder containing the bank statements to Dean’s desk and pulled out the previous month’s statement.

Ahmad looked at the list of cleared Cheques on the bank statement, searching for the check number

for the “void” Cheques. She froze when she saw one of the check numbers listed as “cleared” on the bank statement with the amount \$300 beside it; she had a dreadful feeling that this day was only going to get worse. Over the course of the morning, Ahmad reviewed the check stubs for the past two years, listing every check number marked “void.” She took the list to the bank. She asked the bank manager if she could get copies of the front and back of the cleared Cheques on her list. The bank manager used the computer to find scanned copies of the front and back of every cheque on Ahmad’s list. None of the Cheques were void; every check had “Dean Brown” as the payee and Ahmad could tell her signature was forged. Ahmad took copies of the forged Cheques and left the bank with a heavy heart.

After sitting in her office and thinking about the situation for an hour, Ahmad realized she was “over her head”; she was not trained to deal with embezzlement! After all, who knew how much more Dean had stolen and how? She called the local police department and made an appointment

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for a detective to meet with her on Monday. On Monday, April 18, 2011, Detective Jeff Etty arrived at NUASA to talk with Ahmad. Ahmad explained to Detective Etty what occurred on the previous Friday, and she showed him the forged Cheques, cheque register, and cheque stubs, bank statements, and the purchase card statement that started the whole thing. Detective Etty asked Ahmad to sign her name to five pieces of paper so he could compare her signature to the check

copies. After comparing signatures, it was obvious to the detective that the Cheques were forged. Detective Etty agreed to investigate further. At Detective Etty's request, Ahmad searched Dean's desk for bank reconciliations, she did not find any, although preparing bank reconciliations was one of Dean's responsibilities. Detective Etty asked Ahmad to access copies of the purchase card statements for the card assigned to Dean but, because she did not know the passwords to look up the statements online, she had to call the bank that issued the card and ask for statements to be faxed to the shelter. Once Detective Etty completed his examination of the documents, it was obvious that Dean had been using his purchasing card to make illegal purchases starting in June 2010. Further, Dean had forged Ahmad's signatures on Cheques he wrote to himself since September 2010. After completing his investigation, Detective Etty could prove that Dean illegally purchased \$736.43 of goods from K-Mart, \$75.98 of goods from Staples, forged three Cheques to pay those bills, and forged another 26 Cheques addressed to himself totaling \$10,758.15. In total, Dean had stolen almost \$12,000 in less than a year _ almost 4% of the gross receipts of the shelter during that time period! When Dean returned to work on Tuesday, April 19th, Ahmad terminated him immediately. She was deliberately vague with Dean about the full cause of his termination; she only said that she was concerned about his handling of the shelter's finances.

On Wednesday, April 20, 2011, at 1:00 pm, Detective Etty conducted a taped, Mirandized¹¹ interview with Dean Brown at the police station. During this interview, Dean stated that "he was an artist and would like to pursue his hobby after he gets through this incident," and that "he is into photography."¹² Detective Etty told Dean that the interview was about the forged Cheques and asked him why he forged the Cheques. "Dean stated that he was scared. Dean stated that he did more than his share of work and covered for other people over there (NUASA) all the time and there were certain things that popped up that I just didn't have the income for and I knew what I was doing was wrong." Dean also stated that "car repairs cropped up (tires at a local tire shop) and everyday expenses." Dean further stated that "he was feeling let down at work and knew what he was doing was wrong ... that after he was fired, he was embarrassed and has not shown his face in town ... that he did not even tell his fiancé' ... that he loved his job and loved working with the residents." Dean admitted that he "made out about 10 to 15ish Cheques to himself and signed Ahmad Yaro's name ... Dean was not certain on the total amount of Cheques that he made out to himself ... (and) that he started writing these Cheques about a year ago." He further admitted that "he started the illegal purchases from K-Mart and Staples and that it was stupid ... that he would pay the illegal purchases from K-Mart and Staples by writing NUASA Cheques and forging Ahmad Yaro's signatures to pay the invoices." Dean admitted that "the food purchases and CDs, DVDs and other items purchased from K-Mart were purchased illegally ... (he) thought there were about one dozen illegal purchases from K-Mart and one dozen illegal purchases from Staples ... (he) would purchase ink cartridges from Staples ... (and) the purchases may have started about a

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year ago." During the interview, Dean stated that "he was so sorry and regretful for what he did ... (he) stated that he was going to beg for mercy from God and who ever."

Table 1. Second Chance Homeless Shelter _ Discussion Questions (Phase I: Background).

Questions Solutions

What clue led Allison to suspect possible The purchase card statement was the first fraud? sign of a problem. Once Allison discovered the odd charges at K-Mart and Staples, where could she begin to look in order to discover the extent of Dean's fraudulent behavior? What additional documents might be helpful? Note to instructor: we use this question as a brainstorming tool to get the students to begin thinking about where/how to look for fraud. Furthermore, we use this question to begin exposing them to the different documents available in the accounting records. Possible student solutions include: purchase card statements, check register (checkbook), bank statements, and deposit slips.

When reviewing the checkbook, how could Allison could examine the check register, Allison identify stolen funds? Are there cancelled checks and voided checks. She additional documents or information that could compare the check register and might be helpful? cancelled checks looking for any discrepancies, suspicious or unusual vendors, and forged signatures. Allison could also review any voided checks to make sure they are actually voided and were not used.

How could Allison determine if Dean has Allison could review the donor tax receipts stolen cash donations which should have mailed to donors each year (As this is a been deposited at the local bank? 501(c)(3) organization, each year the Shelter would send tax statements to donors detailing their contributions for the past year.). The shelter should be sensitive to any donor complaints of discrepancies between their records and

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the tax receipt they received. Dean could have been taking donations and then altering the donor receipts to hide his theft. As a proactive measure, the donor tax receipts could include a note: "Please verify that all donations to Second Chance

Homeless Shelter are accurate and report any discrepancies directly to Allison Wright at 800-XXX-XXXX."

How can Allison evaluate Dean's use of Each month, Allison could review Dean's his purchase card? Are there additional purchase card statements and match each documents or information that would be purchase on the statement to the original helpful? store receipts. In this exercise, Dean haddisposed of all purchase card statements related to his card (remember, he wasresponsible for opening all mail and filing the statements?). If she was concerned about Dean's behavior, Allison could contact the purchase card vendor directly for copies of past purchase card statements to review. On an ongoing basis, Allison could review the purchase card activity online.

Q AND ANSWERS

Table 2. Second Chance Homeless Shelter _ Discussion Questions (Phase II: Discovery).

Questions Solutions

Using the fraud triangle framework, (a) Incentive/Pressure The perpetrator identify the components of fraud found in often faces outside financial pressure this exercise: which they

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feel they cannot disclose. For example, in this exercise, Dean faced the unexpected "car repair." In this exercise, our review of court documents, as well as undocumented conversations with those familiar with the case, revealed that Dean had an alcohol problem. After the fraud was discovered, Dean participated in professional treatment services. We discuss this additional factor with students, as well as additional examples (drug use, gambling, sick relative). It is also important to indicate that the financial "problem" can be as simple as greed.

(b) Opportunity The second component of

the fraud triangle results from environmental factors that enable the fraudster to alleviate their financial problem(s) via embezzlement. The most common environmental factor, which provides the opportunity to steal, is a lack of strong internal controls. In addition to a lack of strong internal controls, high levels of trust can also contribute to the opportunity for theft to occur. When teaching this exercise, we typically reserve a discussion of internal controls until the next question; here, we direct students to consider the factors that lead to a lack of internal controls such as:

(1) Lack of financial acumen on the organization's board of directors. (2) Lack of employees with sufficient accounting background including experience in designing and enforcing internal controls. (3) Lack of internal or external audits. (4) Lack of financial resources, which often lead to time constraints or shortcuts. (5) Donation revenue (from non-reciprocal transactions) is easier to steal than

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revenue received in exchange for goods or services (reciprocal transaction) (Ventura & Daniel, 2010).

(c) Rationalization The final component of the fraud triangle is justification by the perpetrator for their actions. Fraudsters often justify their actions with a sense of entitlement to funds (they are overworked), or the rationalization that they are simply "borrowing" the funds (Ventura & Daniel, 2010). They also rationalize that the theft is not causing anyone physical injury. We direct students to the following direct quote (included in the exercise) from Dean indicating his

rationalization: “he did more than his share of work and covered for other people over there (Second Chance) all the time.”

How did Second Chance Homeless Lack of internal controls over the financial Shelter provide the opportunity for Dean to processes of the organization provided the steal from the organization? opportunity for Dean to steal. At this point, we provide a basic definition loosely based on the COSO framework (COSO, 2013) of internal controls as processes and procedures designed to protect an organization from misreporting and to protect its financial assets. Second Chance lacked two specific internal controls that allowed Dean to embezzle.

(a) Segregation of duties: Dean had authority to collect donations and deposit donations. Dean was also allowed to use the purchase card and subsequently received, reviewed, and filed purchase card statements. Dean also had custody of the check register and performed the bank reconciliation.

(b) Reconciliation/Review: Although checks required Allison’s signature, Dean

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was able to circumvent this internal control due to inadequate review. The checkbook was not reviewed on a monthly basis, nor was Allison performing (or even reviewing) monthly bank reconciliations. In addition, no one was reviewing the monthly purchase card statements.

What steps could Second Chance Implementing the internal controls outlined Homeless Shelter take to prevent future in the prior question would allow Second cases of embezzlement? Chance to prevent, or at minimum, more quickly detect embezzlement. We encourage students to think creatively about the internal controls discussed in the prior question, allowing them to design their own methods for segregating duties (e.g., two individuals must deposit

donations, monthly bank, and purchase card reconciliations must be performed by someone who does not have access to the checkbook or have a purchasing card, etc.). This allows the students to gain an appreciation for simple ways to integrate internal controls into an operational process without the need for expensive solutions. During this discussion, we attempt to point out any obvious flaws in their suggested solutions and challenge them to design a better system of internal controls.

Do you think the punishment fit the crime? Again, no predetermined solution is Beyond the loss of funds, what additional possible for this question, but students problems could Second Chance typically engage in interesting dialogue Homeless Shelter face as a result of related to this question. An important Dean's fraud? consequence could be that donors would lose confidence in Second Chance's fiscal responsibility which could result in a loss of donations and, therefore, impair Second Chance's ability to achieve its service-oriented goals.

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